

HOUSE PRICE PREDICTION USING LINEAR REGRESSION

1. Objective

The objective of this project is to build and evaluate a Linear Regression model for predicting house prices using the California Housing dataset. This project helps understand the complete machine learning workflow, including data loading, exploration, preprocessing, model training, prediction, and performance evaluation.

2. Introduction

Machine Learning is a branch of Artificial Intelligence that enables systems to learn from data and make predictions without being explicitly programmed.

Linear Regression is a supervised machine learning algorithm used to predict continuous numerical values by finding the relationship between input variables and output variables.

In this project, the California Housing dataset is used to predict median house values based on different housing features.

3. Dataset Description

The dataset used is the California Housing dataset provided by scikit-learn

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The dataset contains housing information collected from California districts.

Features used:

- MedInc – Median income
- HouseAge – Median house age
- AveRooms – Average number of rooms
- AveBedrooms – Average number of bedrooms
- Population – Population in the area
- AveOccup – Average house occupancy
- Latitude – Geographic latitude
- Longitude – Geographic longitude

Target variable:

- MedHouseVal – Median house value

This dataset contains:

- 20,640 records
- 8 input features
- 1 target variable

4. Tools and Technologies Used

The following tools were used:

- Python
- Jupyter Notebook
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn

5. Methodology

Step 1: Data Loading

The California Housing dataset was loaded using scikit-learn and converted into a pandas DataFrame.

Step 2: Data Exploration

Exploratory Data Analysis (EDA) was performed to understand the dataset.

The following checks were done:

- Display first few rows
- Check dataset information
- Statistical summary
- Missing value analysis
- Feature correlation analysis

Step 3: Data Preprocessing

Input features and target variable were separated.

- X = Input features
- y = Target variable

The dataset was split into:

- 80% training data
- 20% testing data

Step 4: Model Training

A Linear Regression model was created using scikit-learn.

The model was trained using training data.

Step 5: Prediction

The trained model predicted house prices using testing data.

Step 6: Evaluation

Model performance was evaluated using:

- MAE (Mean Absolute Error)
- RMSE (Root Mean Squared Error)
- R^2 Score

6. Visualizations

The following visualizations were created:

Histogram

Used to understand data distribution.

Correlation Heatmap

Used to identify relationships between features.

Actual vs Predicted Scatter Plot

Used to compare actual and predicted house prices.

Residual Plot

Used to analyze prediction errors.

7. Results

- MAE: 0.53
- RMSE: 0.74
- R^2 Score: 0.57

Interpretation

- The MAE value shows the average prediction error.
- RMSE indicates how much the predicted values differ from actual values.
- R^2 score shows how well the model explains the variation in house prices.

The model achieved reasonable performance for a basic regression model.

8. Future Improvements

Model performance can be improved by:

- Feature engineering
- Removing outliers
- Feature scaling
- Hyperparameter tuning

9. Conclusion

In this project, a Linear Regression model was developed to predict house prices using the California Housing dataset.

The complete machine learning workflow including data loading, preprocessing, visualization, training, prediction, and evaluation was successfully implemented.

The model produced acceptable results and provided a strong foundation for understanding machine learning model development.

10. References

- Scikit-learn Documentation
- California Housing Dataset
- Python Documentation